

Characterization of Rotary Atomizer Spray Geometry Under the Influence of a Propeller

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Recent advancements in aerial spraying have driven a shift toward rotary atomizers on unmanned aerial vehicle platforms because they produce uniform droplet sizes at low flow rates with minimal clogging. However, the interaction between rotary atomizer plumes and UAV propeller wakes remains poorly understood. This study uses an optical spray patternator to characterize the three-dimensional spray geometry of a rotary atomizer operated beneath a single 17×7 APC propeller and a dual propeller configuration in the University of Dayton Low Speed Wind Tunnel Laboratory. Intensity based reconstructions quantify changes in spray shape, diameter, and droplet concentration as functions of nozzle rotational speed and distance from the nozzle. A single propeller contracts the natural umbrella shaped plume into a column but causes the high-intensity core to erode rapidly, producing a fragmented spray and a diffused low-density halo. The dual propeller configuration generates a stronger, more uniform downwash that preserves a broad cylindrical high-intensity core, reduced low droplet density zone, and keeps the energetic region closer to the spray center. These results show that multi-propeller flow fields can stabilize rotary atomizer sprays and improve the distribution of deposition relevant droplets in UAV spraying applications.

Nomenclature

W	Weight of the drone	V_0	Rotor inflow velocity
n	Number of propellers	D	Discharge Coefficient
ρ	Fluid density	Re	Reynolds Number
A	Rotor disk area	$D_{Orifice}$	Nozzle Orifice Diameter
C_d	Drag Coefficient	$A_{Orifice}$	Nozzle Orifice Area
$V_{Nozzle Exit}$	Droplet Exit Velocity from Nozzle		Volume diameter with X% of liquid
$V_{Propeller Exit}$	Exit Velocity from Propeller	D_{Vx}	atomized into droplets below this value
Q	Volumetric Flow Rate	R_{prop}	Radius of the propeller(s)
GPM	Gallons Per Minute		

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I. Introduction

In recent years, the use of unmanned aerial vehicles (UAVs) in agriculture has evolved significantly, particularly with the integration of rotary atomizers for crop spraying. These atomizers, also referred to as centrifugal or spinning-disc atomizers, have gained traction as an alternative to conventional hydraulic nozzles in UAV applications due to their unique droplet formation mechanisms and adaptability to varying field conditions. Rotary atomizers generate droplets by imparting centrifugal forces on a liquid film, resulting in a narrower droplet size distribution and reduced susceptibility to clogging compared to flat fan nozzles [1,2]. Several studies have highlighted the advantages of rotary atomizers in aerial agricultural applications. Ranabhat and Price [3] compared centrifugal and hydraulic nozzle configurations and found that rotary atomizers provided comparable swath widths (~5 m) with fewer nozzle units, suggesting more efficient spray coverage. Moreover, rotary systems allow for real-time droplet size control by modulating disc speed, enabling on-the-fly adjustments to environmental conditions. Chen et al. [4] demonstrated that increasing rotational speed of the atomizer led to finer droplets, whereas higher flow rates or fluid surface tension slightly increased droplet sizes. Importantly, this control over droplet characteristics makes rotary atomizers particularly suitable for precision spraying, where reducing off-target deposition is a priority.

From a design standpoint, rotary atomizers are well-suited for UAV platforms due to their compact structure and flow rate scalability. Yang et al. [5] evaluated cage-type rotary atomizers with varying diameters and showed that larger cage diameters produced slightly smaller droplets while maintaining stable operation across a wide range of flow rates. Their study suggests that optimizing atomizer geometry can further reduce drift and improve deposition uniformity in UAV spraying systems. Other works have similarly found that geometric refinements such as edge serration and flow channeling can help sustain uniform atomization across environmental conditions [6,7].

Droplet size remains a critical parameter in spray drift potential. Numerous studies agree that droplets below 140 μm are particularly prone to off-target movement [8,9]. Rotary atomizers, by producing a tight droplet size spectrum, reduce the number of extremely fine droplets, thus minimizing evaporative loss and atmospheric drift [4,5,10]. However, if operated at excessive speeds, they can still generate a high proportion of droplets below 100 μm , necessitating appropriate selection of operating parameters and formulation additives. This balance between coverage and drift control is a key motivation behind tuning atomizer behavior to the ambient airflow environment, particularly in UAV-based applications.

The influence of UAV rotor airflow, or propeller wake, on droplet behavior has also received significant attention. Tierney et al. [11] and Tierney and Gunasekaran [12] performed laser diffraction and high-speed shadowgraph experiments to examine spray evolution in rotor wakes using flat fan nozzles. Their results showed that propeller-induced turbulence widened the spray plume and shifted the droplet size distribution toward larger values due to the removal of finer droplets by upward recirculation. Importantly, their results revealed that conventional point-based droplet measurement techniques like laser diffraction are insufficient to capture the full spatial distortion caused by rotor wakes, motivating the need for methods that measure full-field spray geometry such as spray patterning.

Characterizing spray geometry becomes particularly crucial when considering rotary atomizers, which rely on symmetric ejection of liquid from a spinning surface. In the presence of complex rotor-induced flows, this symmetry is likely disrupted, potentially leading to uneven spray distribution, plume skewing, or loss of fine droplets to drift. Yet, most prior studies have focused on pointwise or average measurements without capturing the spatial footprint of the rotary atomizer under rotor influence. Our previous work [12], [13] demonstrated that even for flat fan nozzles, the spray footprint changed dramatically in the presence of one or more propellers. Given that rotary atomizers produce a more finely controlled yet aerodynamically sensitive plume, it stands to reason that characterizing their geometry using a spatially resolved technique like a spray patternator is essential for understanding the combined effect of disc rotation and rotor-induced turbulence.

This hypothesis is supported by recent CFD and experimental efforts. Carreño Ruiz et al. [14] modeled droplet dispersion under UAV rotor flow during forward flight and found that turbulence from the rotors significantly influenced ground deposition patterns. Matsushita [15] used large-eddy simulations to study rotary atomizer sprays under UAV airflow and emphasized the importance of local velocity fluctuations in modifying spray paths. Similarly, studies by Qiu et al. [16] and Li et al. [17] confirmed that nozzle placement relative to rotors could cause asymmetric deposition patterns even with constant disc speed. These findings align with the field observations by Teske et al. [18], who noted erratic coverage and off-target drift under varying wind and flight conditions. In laboratory studies, Zhang et al. [19] and Wei et al. [20] highlighted that multi-rotor UAV configurations create overlapping wake structures, leading to complex flow interactions that exacerbate plume distortion. Meanwhile, Yang et al. [21] and Zande et al. [22] showed that under certain airflow regimes, droplets from rotary atomizers were recirculated into the rotor system,

further modifying the spray angle and trajectory. This interaction not only impacts coverage and uniformity but may also affect nozzle wear and droplet evaporation dynamics.

Together, these findings underline the need for integrated design approaches that consider both nozzle performance and aerodynamic interactions. Optimizing rotary atomizer design and operating conditions in tandem with UAV flight dynamics offers the best potential for improving spray efficacy and minimizing environmental impact in aerial agriculture. Characterizing spray geometry using a patternator under propeller wake conditions offers a practical and replicable means of gaining such insights and identify optimal placements, orientations, and operational settings for real-world drone spraying scenarios.

II. Experimental Setup

A. Spray Test Stand

All experiments were performed in the University of Dayton Low-Speed Wind Tunnel Laboratory (UD-LSWT) using the propeller-spray test stand shown in Figure 1. The setup features one or two APC 17x7 propellers mounted on parallel linear traverses, each capable of 1 meter of travel. Custom pipe fittings were used to mount standard agricultural nozzles at adjustable transverse and downstream positions within the propeller wake. More details regarding the test stand can be found in [12].

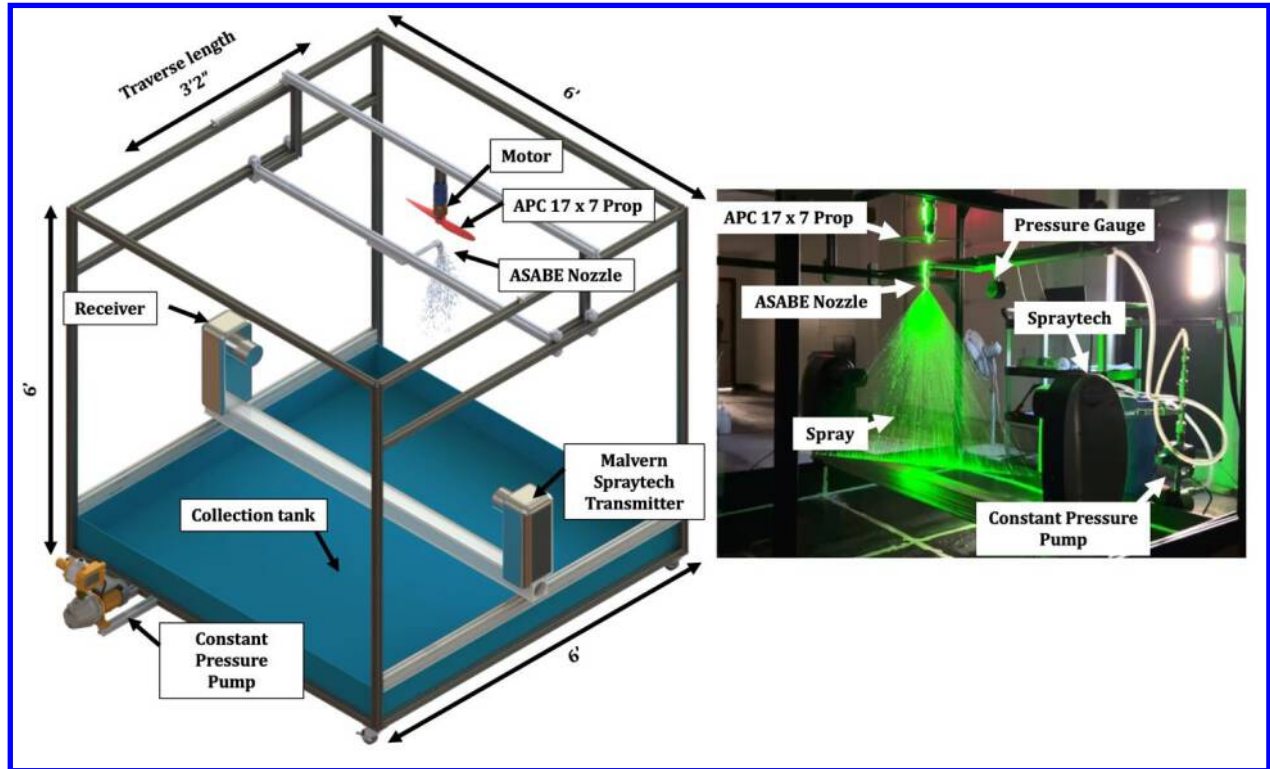


Figure 1 Propeller-spray test stand used for experiments. (Left) Schematic showing the setup with key components, including APC 17x7 propeller, ASABE nozzle, and Malvern Spraytec laser diffraction system. (Right) Photograph of the operational setup, highlighting spray visualization.

B. Rotary Atomizer

The rotary atomizer used in this study was the Micronair Micromiser 12 [23], shown in Figure 2. It is a compact, precision atomizer designed to produce droplets in the 60–300 μm VMD range. Spray droplets are generated by a rotating toothed disc, with droplet size governed primarily by the disc's rotational speed. The disc is driven by a low-voltage brushless motor with closed-loop control electronics that maintain constant RPM over the operating range of liquid flow rates. The motor is supplied by a 24 V DC power source and includes a speed control interface that

allows RPM adjustment through an external potentiometer. The atomizer was installed vertically with the disc facing downward and connected to a liquid feed line and vent tube in accordance with the manufacturer's guidelines.

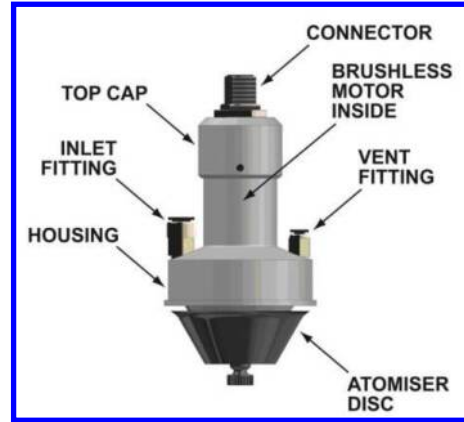


Figure 2 Mirconair Micromiser 12 rotary atomizer

The spray diameter from the atomizer varies with rotational speed, as shown in Figure 3. At the lower end of the RPM range, the rotary atomizer spinning at 2800 RPM produces relatively coarse droplets with diameters of approximately $220\ \mu\text{m}$. At the higher end, 4500 RPM generates droplets of about $150\ \mu\text{m}$, and droplets of this size or smaller are more prone to drift. The RPM of the rotary atomizer was set using a custom-made RPM controller and verified by reading the encoder output on an external oscilloscope. The potentiometer on the controller was adjusted until the live frequency on the oscilloscope matched the desired RPM.

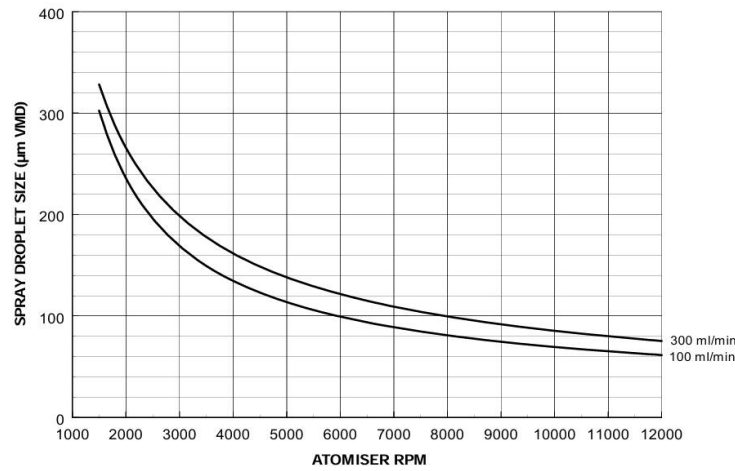


Figure 3 Atomizer RPM, fluid flow rate, and spray droplet size relationship chart [22]

B. Propeller

All nozzles were tested at a baseline of 0 propeller RPM and at 4900 RPM. Thrust, propeller RPM, and throttle settings were determined using the propeller performance tests conducted at UD-LSWT, as reported by Cai et al. [17]. The exit velocity from the propeller was estimated using Equation 1, derived from actuator disk theory [18]:

$$V_{\text{Propeller Exit}} = \sqrt{\frac{2T}{\rho A}} \quad (1)$$

Where $V_{\text{Propeller Exit}}$ represents the exit velocity by the propeller, T represents the measured thrust, and A represents the rotor disk area calculated from the propeller diameter. Table 1 summarizes the chosen throttle settings, the corresponding thrust, and the estimated exit velocities. The thrust values at 3900, 4900, and 5900 RPM for the APC

17x7 propeller produced exit velocities similar to those of DJI UAVs, which generate induced velocities ranging from 5 to 15 m/s.

Table 1 Thrust, throttle settings, propeller RPM, and estimated exit velocities of the APC 17x7 propeller

Thrust (N)	Throttle Setting (%)	RPM	Density (kg/m ³)	$V_{\text{Propeller Exit}}$ (m/s)
20.53	40	3900	1.18	7.5
23.62	50	4900		8.3
35.63	65	5900		10.1

The effect of the propeller on the rotary atomizer can be seen in Figure 4. In the absence of external airflow, the rotary atomizer ejects fluid radially from the disc edge, producing a hollow-cone structure that naturally expands outward. This expansion is governed by centrifugal forces and aerodynamic drag, which depend on droplet size. The resulting spray plume spreads laterally and falls slowly as a mist. With the introduction of the propeller, the airflow fundamentally alters this natural spray geometry. The propeller wake accelerates the fluid downward, creates a strong vertical velocity gradient, and can impose asymmetry in the droplet distribution. In this study, the rotary atomizer and propellers rotate in the same direction (clockwise). Under the influence of the propeller, the cross-sectional area of the spray decreases significantly. Without the propeller, the diameter of the spray from the rotary atomizer is approximately 6 ft, whereas the propeller-induced flow reduces this diameter, with the extent of contraction depending on the propeller RPM.

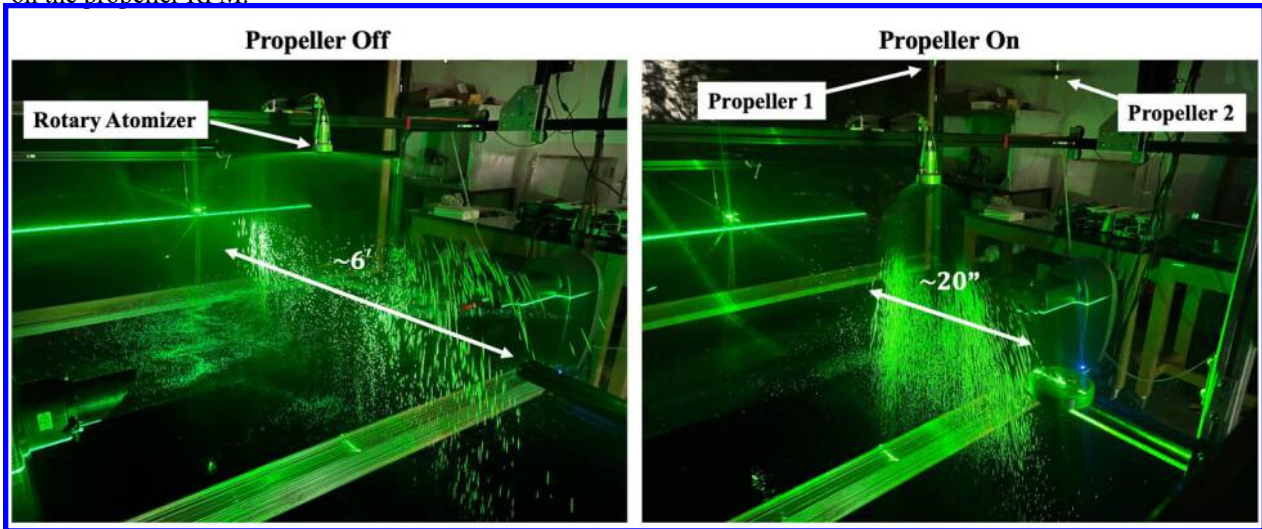


Figure 4 Flow visualization of the rotary atomizer spray geometry with and without the influence of the propeller

C. Spray Patternator

The spray patternator employed a 1 W, 532 nm laser (OX Lasers) oriented perpendicular to the plane of the spray to illuminate the cross-sectional geometry of the nozzle plume, as shown in Figure 5. Multiple laser configurations were evaluated to balance intensity and coverage, and the final setup used two plano-concave lenses (focal lengths -20 mm and -7 mm) to expand the beam and provide adequate illumination across the spray width. Several camera systems were tested to meet requirements for field of view, depth of field, and aperture.

The final configuration used an iPhone 17 Pro Max with a 48 MP sensor and a 24 mm equivalent $f/1.78$ lens to capture the backscattered light from the illuminated droplets. Both the laser and camera were mounted on a vertical traverse, which enabled sequential imaging of spray cross-sections at various heights below the nozzle. A sensitivity analysis was conducted to identify the optimal camera angle for maximizing backscatter intensity. The test matrix for spray geometry characterization is presented in Table 2, and the corresponding imaging planes are shown in Figure 6. For each test, the patternator recorded a 10 s video at 24 fps, beginning 10 in. below the nozzle and continuing in 1 in. increments down to 30 in. Each nozzle was tested at multiple RPMs for two flow-interaction configurations, one with a single propeller mounted above the atomizer and a second with two propellers, where the second propeller was

positioned to the left of the first at a hub-to-hub spacing of 18 in. In both the configurations, the propeller RPM was held constant at 4900 RPM.

All tests were conducted with tap water at a maximum volumetric feed rate of 300 mL/min, supplied by a peristaltic pump through approximately 3 ft of 6 mm tubing. Each test used either a single propeller located directly above the rotary atomizer or two propellers, with the second positioned 18in. to the right of the center propeller and separated by a 1 in. gap. To ensure that observed differences in plume behavior and geometry arose solely from changes in nozzle RPM and configuration, all tests were performed with the same fluid medium, lighting environment, camera exposure settings, and vertical spacing. As a result, variations in the measured plume behavior can be linked directly to changes in nozzle RPM and propeller configuration.

Table 2 Test Matrix used Spray Patternator Tests

No. of Propellers	Rotary Atomizer RPM	Flow Rate (fl oz/min)	Field of View (in)	Measurement Location from Nozzle (in)	Propeller RPM
1	2800, 3000, 3500, 4000, 4500	10.14	57.75" x 48.13"	10" to 30" in 1" increment	4900
2	2800, 3000, 3500, 4000, 4500	10.14	72.84" x 44.92"	10" to 30" in 1" increment	4900

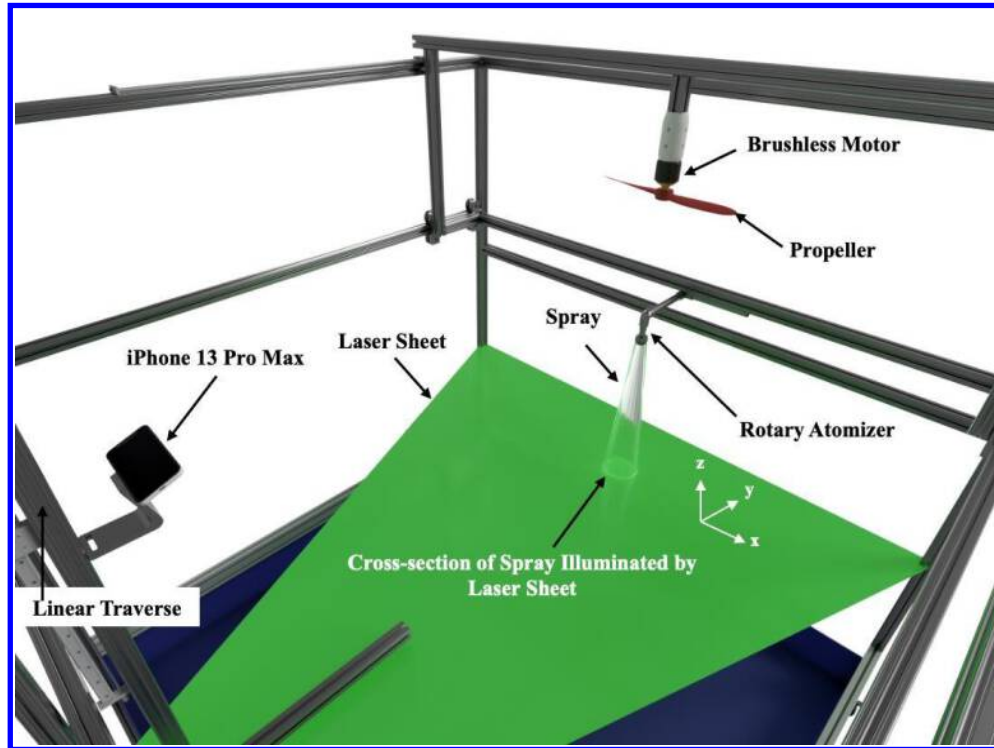


Figure 5 Spray patternator setup showing the alignment of the 532 nm laser, rotary atomizer nozzle, and iPhone 17 Pro Max camera to capture the cross-sectional geometry of the illuminated spray

D. Workflow for Processing Data from Patternator

The workflow used to generate a 3D reconstruction of the nozzle spray geometry is summarized in Figure 7. Because the camera was positioned at an angle relative to the laser light sheet, perspective correction was required to mitigate parallax errors. Images before and after correction are shown in Figure 7. After calibrating the field of view,

the laser sheet and camera were traversed to multiple vertical locations along the spray. At each location, images were recorded at 24 fps for 10s and averaged to obtain a single representative image. Each averaged image was then mapped through the perspective correction and interpolated onto a common grid to reconstruct the 3D spray geometry.

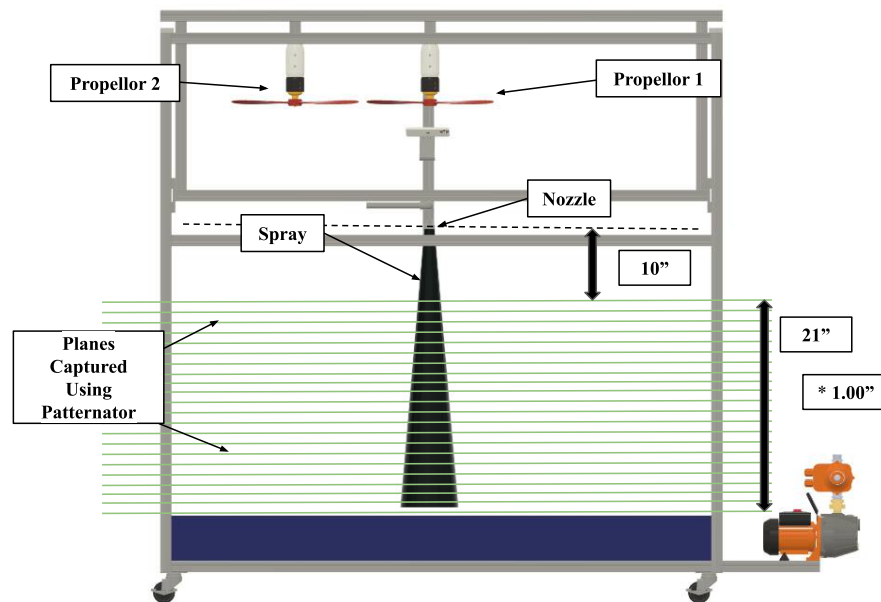


Figure 6 Schematic showing the planes captured using the spray patternator, starting 10 inches from the nozzle and progressing in 1-inch increments

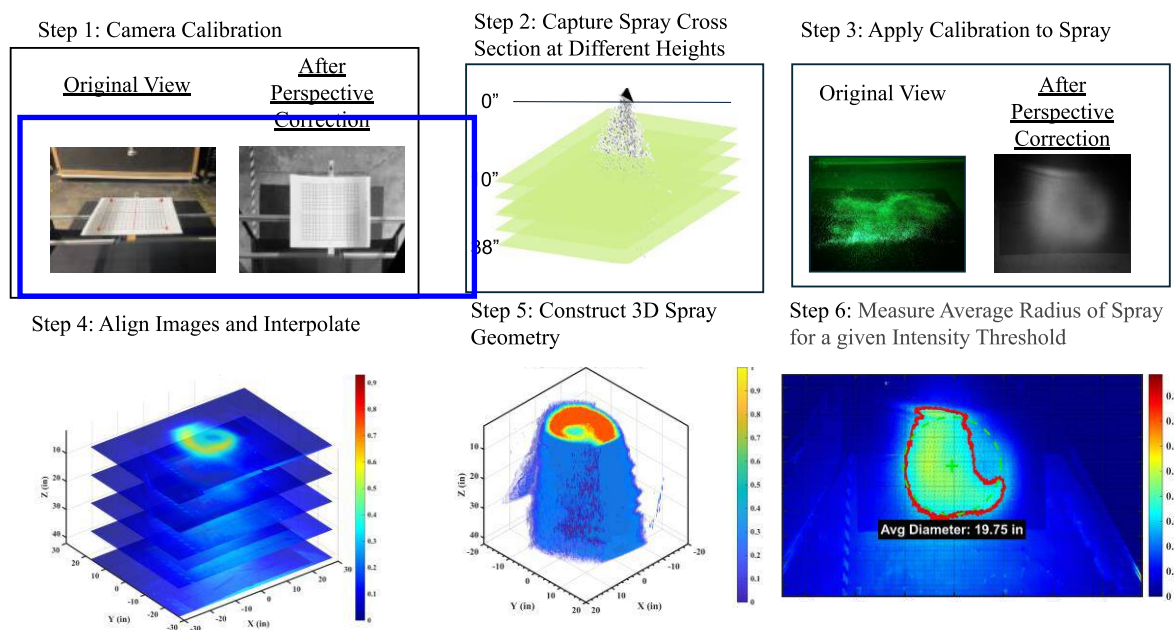


Figure 7 Workflow for processing patternator data, including raw and perspective-corrected images (top row), planes captured at different vertical positions (middle column in first row), and 3D reconstruction of nozzle spray geometry (bottom row)

III. Results

A. Spray Geometry

The 3D reconstructions of the spray geometries were visualized using iso-surfaces of normalized intensity values from 0.9 to 0.1 for all atomizer configurations, as shown in Figure 8. Transparency was applied to the iso-surfaces, with higher transparency near the spray edges and lower transparency near the high-intensity core. This approach improves visualization of regions with higher droplet density within the spray. The first row corresponds to spray volumes exposed to a single propeller, and the second row represents cases with two propellers. The intensity contours in Figure 8 reflect fluid illuminated by the laser and captured by the camera as droplets cross the laser plane, and are indicative of local droplet concentration. High-intensity regions mark denser droplet clusters, while the lighter outer surfaces indicate lower concentration near the periphery. This visualization approach makes it possible to identify core structure, dead zones, and plume skewing under the influence of the propeller wake and downwash.

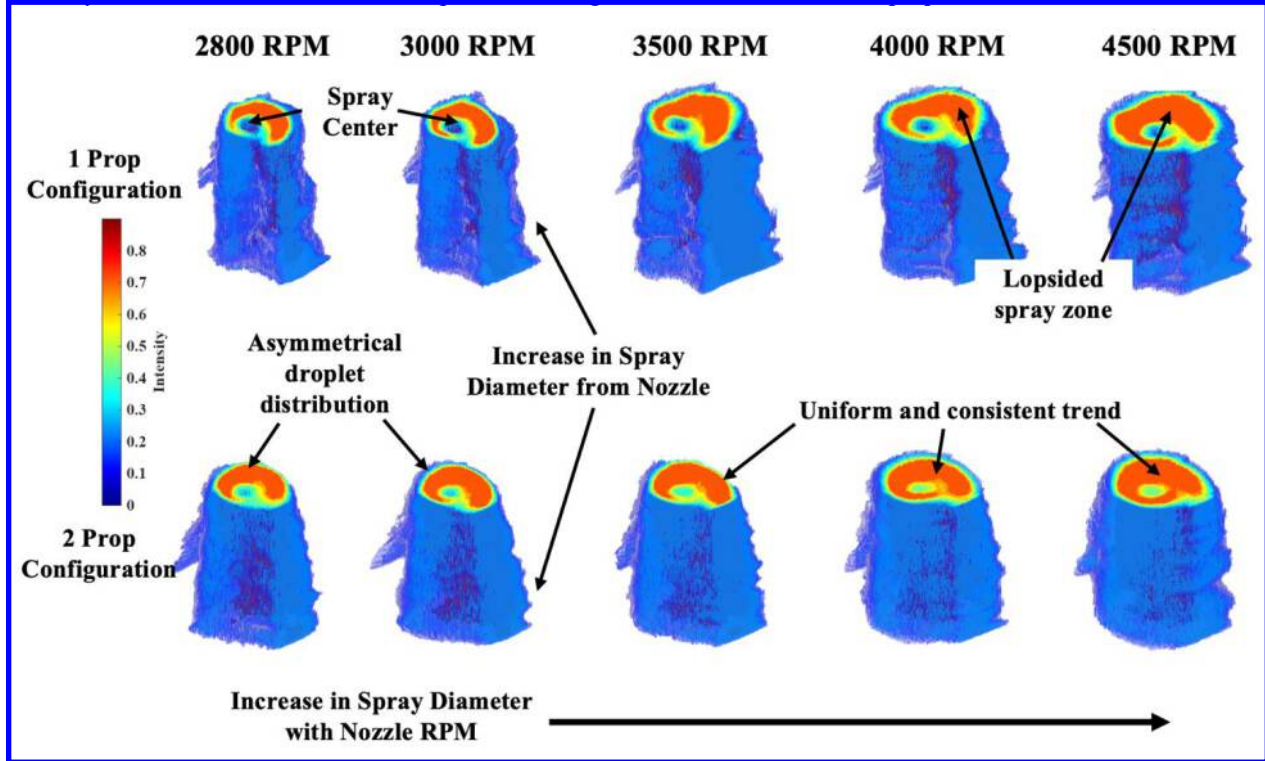


Figure 8 3D reconstructions spray from a rotary atomizer across different RPM and under the influence of either one or two propellers.

As seen in the flow visualization in Figure 4, the spray from the atomizer resembles an umbrella in the absence of a propeller. With the introduction of a propeller, the spray diameter contracts as the propeller wake pushes the fluid downward, which is evident in Figure 8. A consistent feature in all spray geometries is the dead zone visible in the center. This region lies directly beneath the rotating atomizer, which propels fluid laterally, so it contains fewer droplets than the surrounding spray. Under the influence of two propellers, the sprays consistently exhibit higher overall droplet density compared with the single-propeller cases at equivalent nozzle RPM. This behavior can be linked to the downwash of the second propeller, which provides additional lateral confinement and limits radial expansion relative to the single-propeller configuration.

Across all spray geometries, regardless of nozzle RPM or the number of propellers, a strong asymmetry in the spray distribution is apparent. The spray volumes tend to congregate to one side in a thicker, denser region. This zone of high density corresponds to the side of the nozzle where the fluid is injected, as indicated by the inlet fitting in Figure 2. It is difficult, however, to isolate whether the propeller action itself modifies this intrinsic asymmetry, even though clear differences appear in the spray geometry when the second propeller is added.

With the second propeller, the lower-intensity side of the spray appears more evenly distributed than in the single-propeller case at the same nozzle RPM. Although the spray still forms an uneven annulus, the two-propeller configuration produces a smaller zone of low-intensity spray when compared to the single-propeller case. And the

high-intensity region is more uniformly distributed and more consistent across nozzle RPMs. The single-propeller configuration shows a slightly more lopsided spray at higher nozzle RPMs, a feature that is less evident with two propellers. This aligns with the observation of a smaller dead zone for the two-propeller cases, in which the spray as a whole is more evenly distributed, even though prominent asymmetry remains. The two-propeller sprays also exhibit a more cylindrical profile and more circular boundaries. Finally, the diameter of the cylindrical spray geometry clearly changes with distance from the nozzle. The variation of spray diameter with vertical distance and nozzle RPM is quantified in the next section.

B. Cross-sectional analysis of Spray Geometry

The 3D reconstruction of the spray reveals changes in the global structure and shape of the spray geometry at different nozzle RPMs and propeller configurations. However, changes in the interior of the spray are also important to characterize, since variations in droplet density and distribution as a function of distance from the nozzle affect the coverage area, droplet driftability, and the efficacy of the spray over the crop canopy. A set of representative cross-sectional images is shown in Figure 9 for different nozzle RPMs and at various distances from the nozzle for both single and dual propeller cases. The contours also highlight an intensity threshold of 0.3 in red.

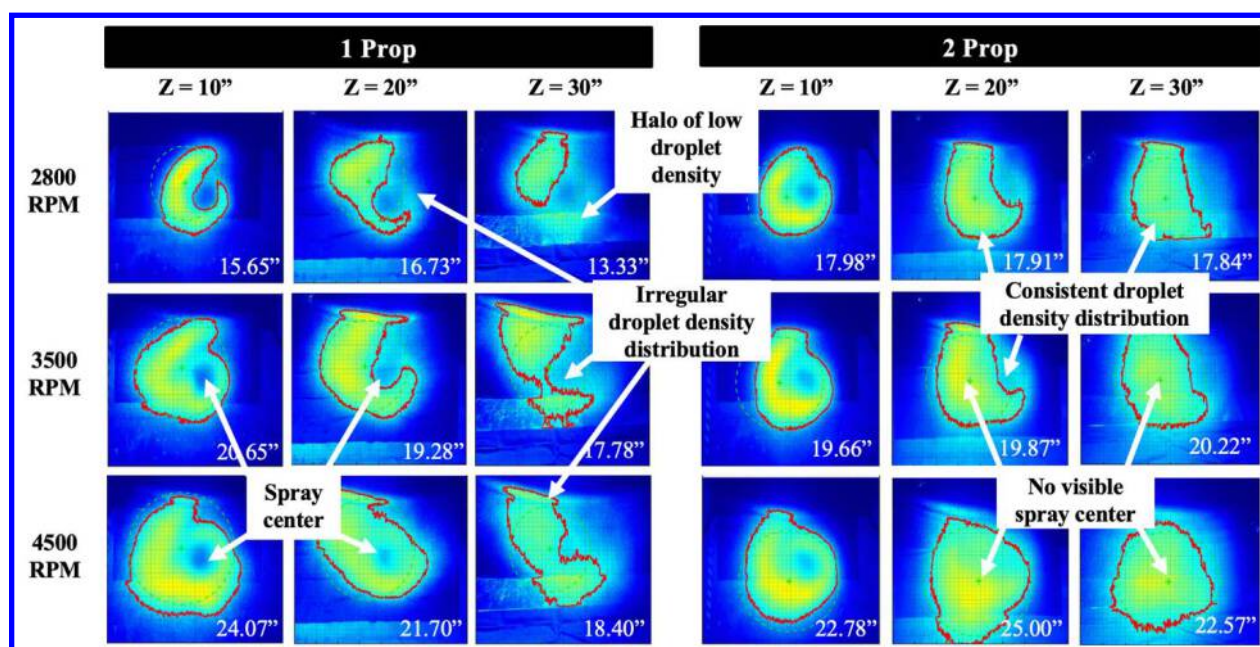


Figure 9 Representative spray cross-sections at three heights and nozzle RPMs for single and dual propellers; red contour marks intensity threshold 0.3 delimiting high-density core.

The high-intensity region enclosed by the 0.3 intensity threshold represents the portion of the spray that contains high droplet density. This region is the most relevant to agricultural spraying applications because it highlights the effective deposition of droplets that are sufficiently concentrated to adhere to and impact crop surfaces. Conversely, the low-intensity halo that surrounds this boundary contributes minimally to deposition and is far more susceptible to atmospheric drift. Analyzing and quantifying how this dense core expands, contracts, or collapses with distance from the nozzle provides insight into geometric spray stability and the retention of deposition-relevant droplets as the spray travels downward.

The shape of the spray in each cross-sectional slice changes noticeably as the plume travels away from the atomizer, especially in the single-propeller configuration. At $Z = 10$ in., all three RPM conditions retain a full, continuous boundary around the high-intensity core, with the previously noted asymmetry and central dead zone still prominent. Although the contours are irregular, the outline still forms a single cohesive shape. At higher nozzle RPMs the threshold boundary is larger. This is due to the fact that at higher RPM, the droplets are smaller, but the number of droplets produced by the atomizer increases because the volumetric flow rate remains constant. This increase in droplet count, combined with higher centrifugal forces, results in a larger spray diameter. By $Z = 20$ in., the shape of this region begins to shift and deform for each case. The boundary contracts and becomes more irregular. Instead of a

high-intensity region that encloses the spray center, the contour pulls to one side and forms elongated, distorted shapes. This indicates that the concentrated portion of the plume is already thinning, even though the full spray, if viewed without intensity filtering, would appear to extend outward and dilate. The strong change in the 0.3 threshold region across these planes shows that a growing fraction of droplets no longer meets the threshold and therefore falls outside the dense core. This effect is more pronounced at $Z = 30$ in., the last Z -slice captured. At this distance the threshold boundary collapses into a small remnant of the original shape and fractures into disconnected regions. The marked reduction in the high density area is evident across all three RPM configurations, although the geometry varies from case to case. Physically, this behavior reflects the decay of the dense central region as droplets diffuse outward and the plume loses cohesion under the single-propeller downwash. These three planes show that with a single propeller mounted directly above the rotary atomizer, the high-intensity core of the spray does not persist far downstream. While the overall plume continues to expand, most droplets have dispersed enough by $Z = 30$ in. that they fall below the 0.3 intensity threshold. In this sense, the shrinking diameter of the threshold boundary indicates a loss of droplet concentration rather than a true narrowing of the spray.

The behavior of the high-intensity region changes considerably and consistently when a second propeller is introduced. At $Z = 10$ in., all three RPM cases show a boundary that is much more uniform than in the single-propeller configuration. Although these sprays are not perfectly circular and the contours remain irregular, they are smoother and more centered than in the single-propeller case. Even at 4500 RPM, where the droplets are finest and most susceptible to the propeller wake, the boundary retains a clear, continuous outline rather than stretching or skewing to one side. This pattern suggests that the second rotor wash provides a stabilizing effect. At $Z = 20$ in., the difference between the single and dual propeller configurations becomes even more distinct. The high-intensity region does not begin to collapse or break apart, but instead remains nearly constant or grows slightly. The threshold contours stay continuous and relatively smooth, with the core still occupying a substantial portion of the field of view. Rather than thinning into disconnected patches, the dense region shifts and stretches within a confined area, which indicates that the dual-prop flow holds the droplets together more effectively as they move downstream. Even the low-density zone that previously marked a clear spray center becomes partially filled by droplets influenced by the second propeller at higher Z locations. By the time the plume reaches $Z = 30$ in., the dense core has weakened, but it still appears as a single coherent boundary in all RPM conditions. Across all three cases, the variation with Z is modest when compared with the strong contraction observed in the single-propeller configuration. The addition of a second propeller slows the rate at which the central region diffuses and keeps a larger fraction of droplets above the intensity threshold even far from the nozzle.

C. Characterization of Spray Diameter

This section examines how the average diameter of the high-intensity core of the spray evolves as it travels downward from the rotary atomizer. Quantifying spray diameter from images is sensitive to the choice of intensity threshold that defines the spray boundary. To capture this dependence, two intensity thresholds, $I_T = 0.2$ and $I_T = 0.3$, were used to extract the spray boundary in each Z -slice obtained from the patternator. A representative example of the thresholding procedure and radius construction is shown in Figure 10, and additional cross sections for different nozzle RPMs and vertical locations are provided in Figure 9 for both the single and dual propeller cases.

Because the spray geometry is asymmetric and irregular, the spray center was manually identified at each vertical location for all cases. The center was defined as the dead zone of minimal particle density directly beneath the atomizer. Once the center was marked and the threshold contours were identified, radial lines were cast through 360° , and the distance from the center to the boundary was measured along each ray. These radii were then averaged and multiplied by two to obtain a representative average diameter for each Z location.

The variation of spray diameter at $Z = 10$ in., denoted as d_{10} , as a function of nozzle RPM is shown in Figure 10 together with the average diameter taken over all 21 Z -slices for both the single and dual propeller configurations. Results are plotted for both thresholds, $I_T = 0.2$ and $I_T = 0.3$. Since $I_T = 0.2$ corresponds to a lower intensity level, the diameters derived from this threshold are larger than those obtained with $I_T = 0.3$. The gap between the two threshold curves provides a measure of the low droplet density halo surrounding the high droplet density region. At higher RPMs the single-propeller configuration tends to show a larger separation between the $I_T = 0.2$ and $I_T = 0.3$ diameters, which indicates that a greater fraction of its footprint is occupied by diffuse, low-density droplets that are more susceptible to drift.

The behavior of d_{10} with nozzle RPM shows an almost linear increase for both configurations and for both thresholds. This trend is consistent with the production of smaller droplets at higher rotational speed and an increased centrifugal force resulting in a larger diameter. Near the nozzle, the single-propeller configuration produces slightly

larger values of d_{10} than the dual-propeller configuration, which indicates a more rapid initial flaring of the spray. In contrast, the average diameter computed over the full vertical extent is larger for the two-propeller configuration at every RPM tested. This difference reveals a clear contrast between near-field and overall behavior. The single-propeller case allows the plume to flare outward quickly then lose its dense core as it moves downstream, whereas the dual-propeller case maintains a broader high-intensity footprint over a much greater vertical distance.

The error bars in the average-diameter plot show that scatter about the mean increases at higher RPM in the single-propeller configuration, which is consistent with the irregular and fragmented cross sections seen in Figure 9. The dual-propeller configuration exhibits comparable or smaller uncertainty, again suggesting a more stable and repeatable spray footprint. From an application standpoint, the larger average high-intensity diameter and improved stability in the dual-propeller case imply a wider band of deposition-relevant droplets over the canopy for the same nozzle RPM.

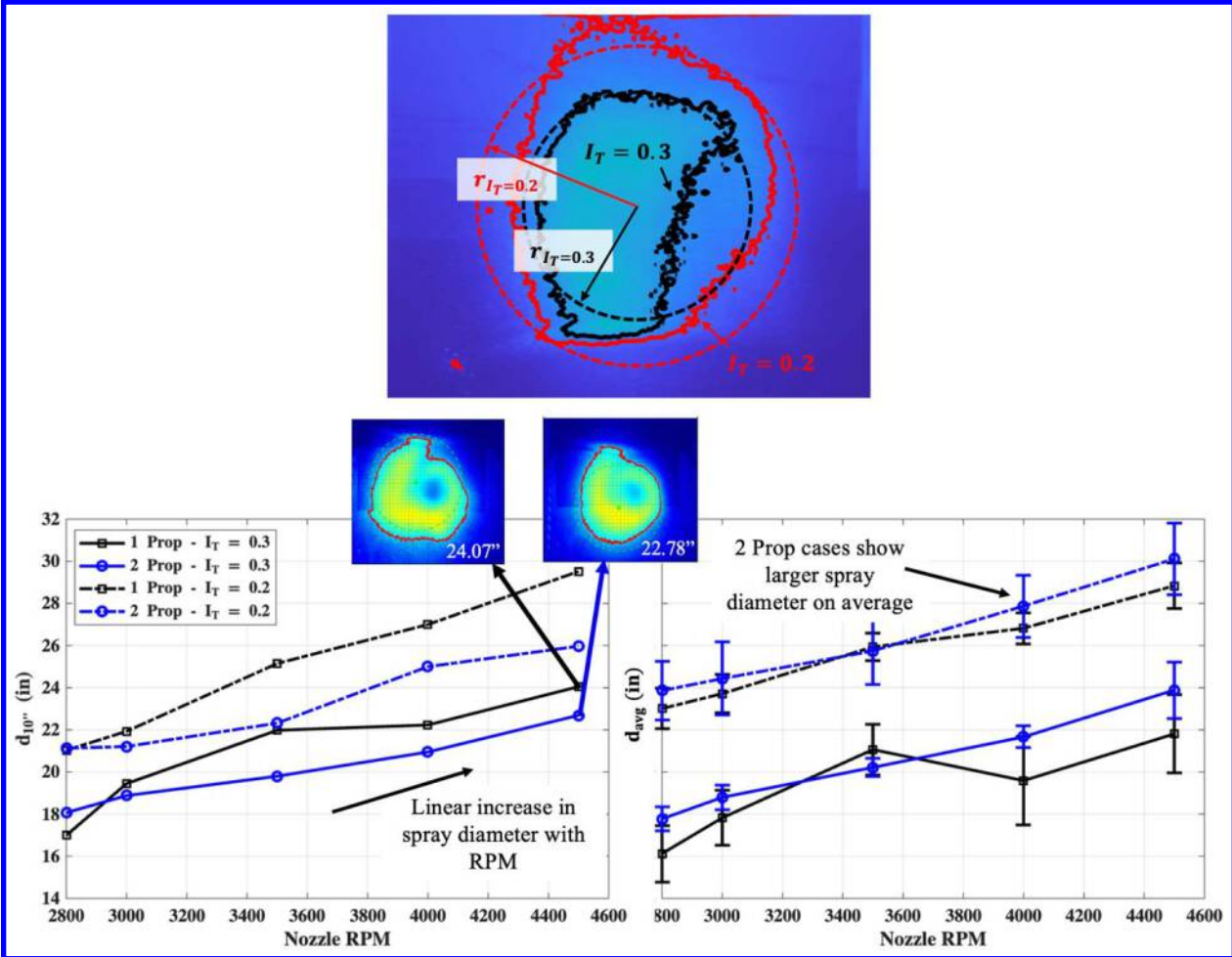


Figure 10 Spray diameters at $Z=10$ in and vertically averaged versus nozzle RPM for two intensity thresholds, comparing single and dual propeller configurations with uncertainty.

The spray diameters normalized by d_{10} for each Z -slice are shown in Figure 11 for both intensity thresholds and for all nozzle RPMs in the single and dual propeller configurations. The vertical axis is also normalized by 10 in., which corresponds to the location of the first Z -slice from the patternator. As seen earlier in Figure 9, the region of high droplet density diffuses as it convects downstream from the nozzle. This behavior is reflected in the curves for the higher intensity threshold. For $I_T = 0.3$, the single-propeller configuration shows a progressive decrease in normalized diameter with increasing Z/Z_{10} . This indicates that the high-density portion of the plume shrinks as the spray travels downward and that a growing fraction of droplets falls below the 0.3 intensity threshold. In contrast, the two-propeller configuration exhibits an almost uniform normalized diameter for the same threshold. The curves cluster

near $d_{I_T}/d_{10''} \approx 1$ over much of the vertical extent, which suggests that the size of the high-density core remains nearly constant with distance. This trend is consistent across RPMs and supports the interpretation that the second propeller provides a stabilizing influence on the dense core of the spray.

The behavior at the lower intensity threshold of $I_T = 0.2$ highlights the evolution of the low-density halo. At this threshold the single-propeller cases show only modest changes in normalized diameter with Z . In several RPM conditions the curves even tilt slightly inward, which implies that portions of the halo decay below the 0.2 threshold as the plume descends. The dual propeller configuration behaves differently. For $I_T = 0.2$ the normalized diameter increases with distance, especially at higher nozzle RPMs. This indicates that the low-density zone surrounding the core continues to expand while still remaining above the detection threshold, so the overall footprint of deposition-relevant droplets grows with downstream distance. These trends help explain why the vertically averaged diameter in Figure 10 is larger for the two-propeller configuration. The single-propeller wake allows the dense core to erode and the halo to thin out so that many droplets either diffuse beyond the measured footprint or become too diluted to register above the intensity thresholds. With two propellers, the combined downwash and lateral confinement keep both the core and the surrounding halo intact over a longer vertical distance. As a result, a larger fraction of droplets remains within a broad, continuous footprint and above even the lower intensity threshold. From a drift perspective this suggests that the dual propeller configuration maintains a wider and more cohesive spray column, which reduces the likelihood that droplets disperse into highly dilute, drift-prone regions outside the primary application zone.

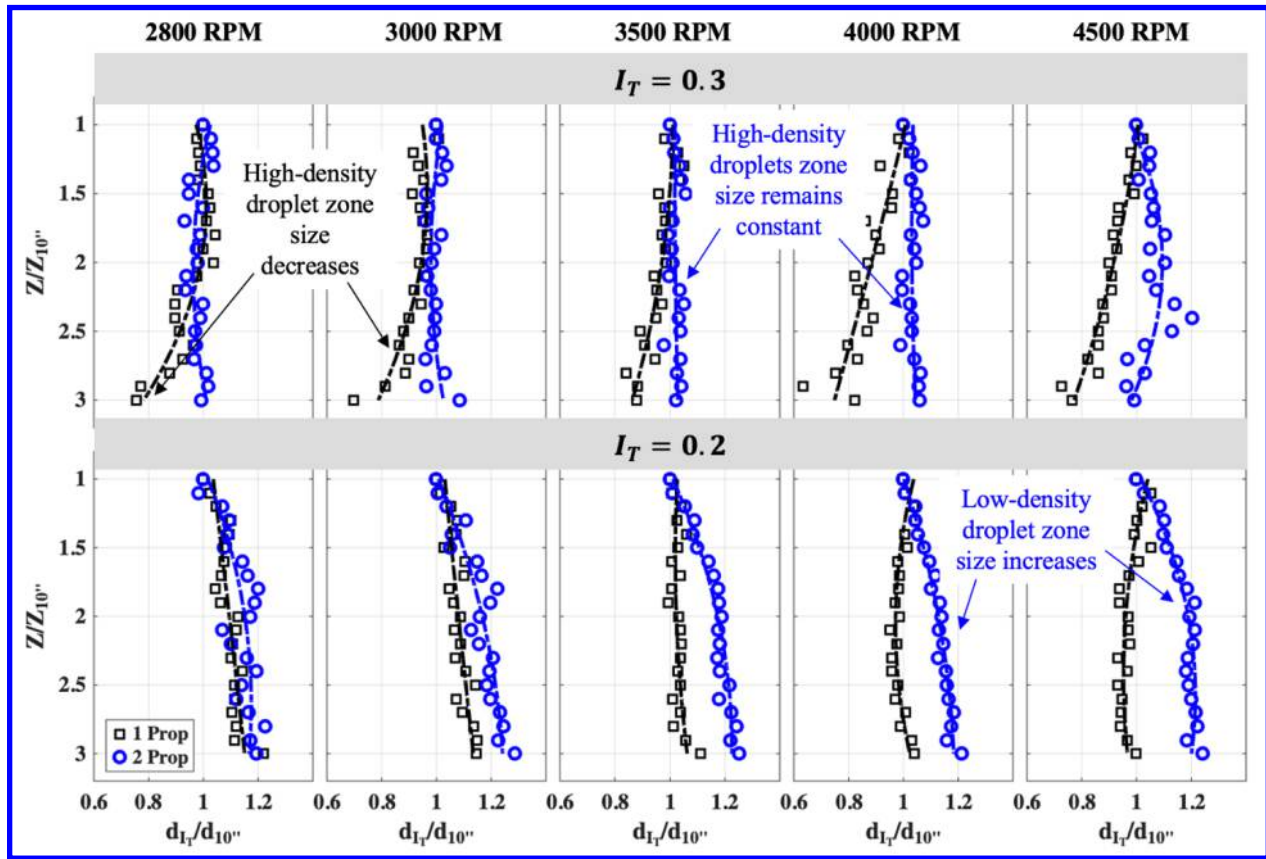


Figure 11 Normalized spray diameter versus height for intensity thresholds 0.3 and 0.2, comparing single and dual propeller configurations and highlighting stabilized core and expanding halo.

The difference between the diameters computed at the two intensity thresholds provides a direct measure of the thickness of the low-density halo that surrounds the dense core. Figure 12 shows this quantity, $\Delta d_{(I_T=0.2-I_T=0.3)}$, as a function of normalized height $Z/Z_{10''}$ for each nozzle RPM and for both propeller configurations. Small values of Δd indicate that the 0.2 and 0.3 contours lie close together, which corresponds to a compact spray where most of the footprint is filled with relatively high droplet concentrations. Larger values of Δd imply that the 0.2 contour

extends well beyond the 0.3 contour, so a substantial fraction of the footprint is occupied by a diffuse halo of lower-density droplets that are more susceptible to drift. For the single-propeller configuration, Δd generally increases with downstream distance across all RPMs. This trend shows that as the plume descends, the high-intensity core shrinks while the lower-intensity halo persists and spreads outward. The growing separation between the two thresholds reflects a transition toward a more dilute and less coherent spray, consistent with the collapse of the 0.3 contour and the strong diffusion seen in Figures 9 and 11. In contrast, the dual propeller configuration maintains smaller values of Δd over most of the vertical extent and for nearly all RPMs, except for the 4500 RPM. The curves for the dual-propeller case do not go beyond $0.2\Delta d$ which indicates that the dense core and the surrounding halo stay more tightly coupled and that the spray footprint remains more uniformly filled with deposition-relevant droplets.

Together with the diameter trends in Figures 10 and 11, Figure 12 reinforces the conclusion that the dual propeller configuration produces a more coherent and stabilized spray column. The second propeller limits the growth of a dilute outer halo and preserves a thicker layer of higher-density droplets across the footprint. From an application standpoint, a smaller Δd implies less of the spray volume is relegated to low-concentration, drift-prone regions, which suggests that the dual propeller arrangement can improve both deposition efficiency and spray containment relative to the single-propeller case.

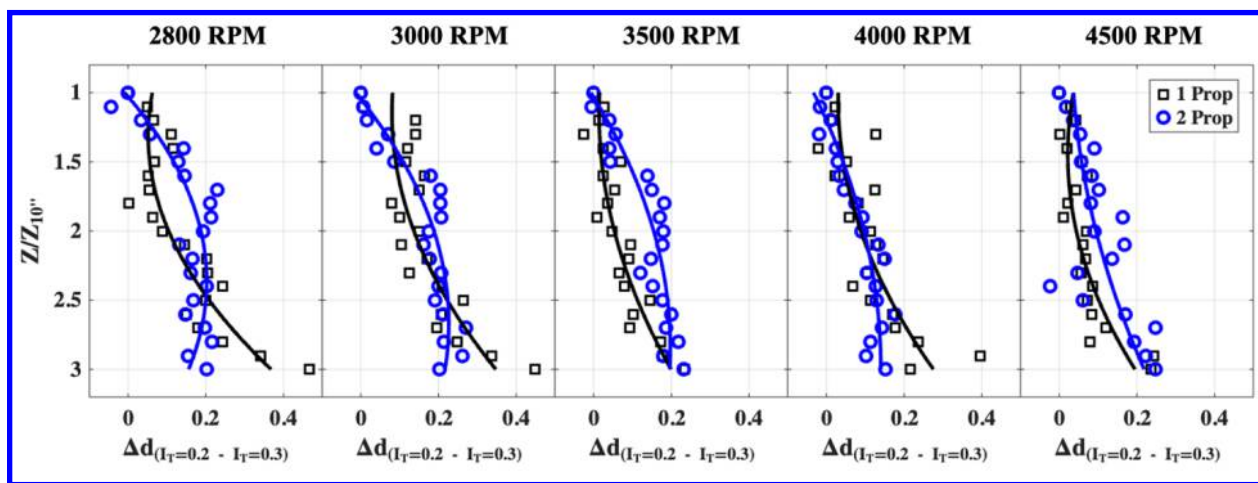


Figure 12 Difference between spray diameters at intensity thresholds 0.2 and 0.3 versus height, showing reduced halo thickness and greater coherence for dual-propeller configuration.

D. Location of Maximum Intensity

The location of maximum intensity in the spray corresponds to the region of highest local droplet concentration, so its path reveals how the energetic core of the plume travels and how stable it is. Large lateral shifts or fragmentation of this peak can lead to nonuniform deposition patterns and greater sensitivity to drift, while a smooth, repeatable path indicates a more controlled and predictable spray.

The first row in Figure 13 shows the spray geometry for the single-propeller configuration. The colored markers trace the location of maximum intensity along the spray as a function of distance from the nozzle. Near the nozzle, the peak intensity lies on the lopsided portion of the plume and remains biased to that side farther downstream, consistent with the inlet-driven asymmetry discussed earlier. As the plume convects downward, the peak locations begin to wander and eventually scatter laterally, especially at higher nozzle RPM. This erratic motion indicates strong dispersion and intermittency in the core, and it aligns with the collapse of the high-intensity region and the growth of the low droplet density halo seen in Figures 9 through 12. In the single-propeller wake, the coherent core survives only over a limited vertical distance before breaking into a set of isolated hotspots.

In contrast, the sprays exposed to two propellers display a much more consistent evolution of the maximum intensity location, as shown in the second row. The markers follow a smooth, continuous path through the spray volume with much smaller lateral excursions, and this trend is similar across all nozzle RPM cases. The core remains biased toward the inlet side, so the intrinsic asymmetry is still present, but the dual-propeller wake organizes this off-center core into a stable and predictable trajectory rather than a wandering hotspot. Viewed together, these trends show that the second propeller extends the range over which a coherent high droplet density zone is maintained and reduces

the magnitude of lateral wandering. This behavior provides additional evidence that the dual-propeller configuration stabilizes the rotary atomizer plume and supports a more repeatable deposition footprint than the single-propeller case.

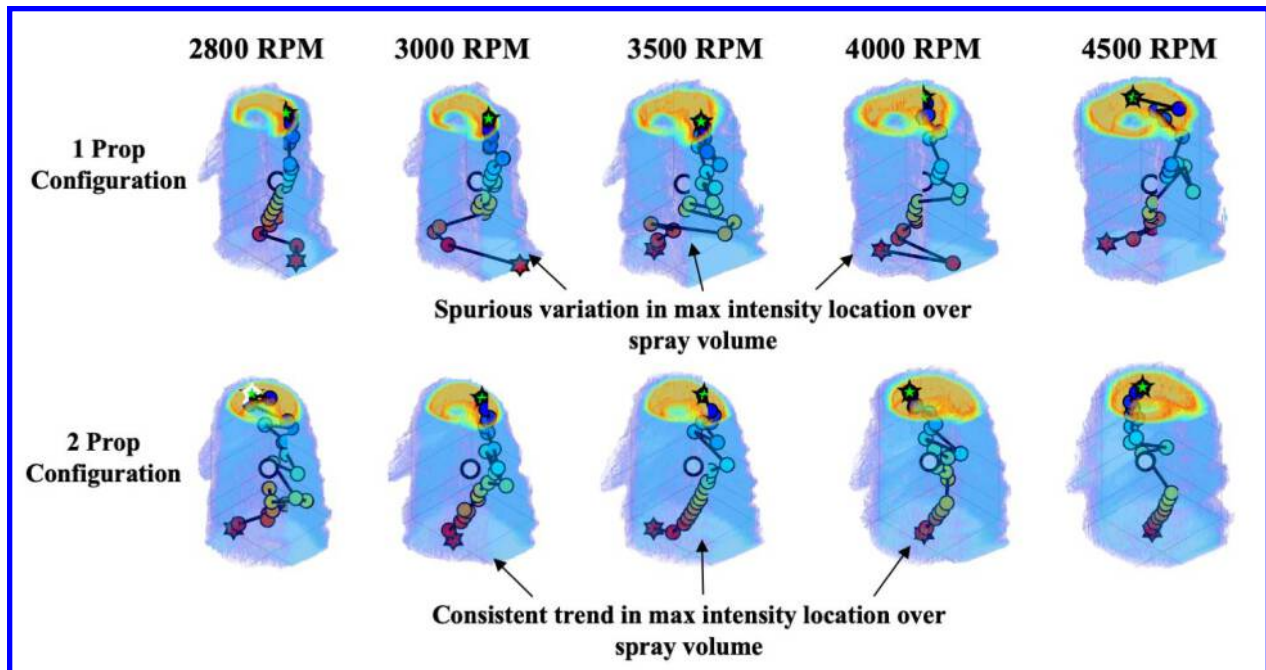


Figure 13 Location of maximum spray intensity for single and dual propeller configurations, showing erratic downstream wandering with one propeller and a stable, coherent core with two.

While Figure 13 shows how the locus of maximum intensity moves through the spray volume, it does not quantify how far that energetic core sits from the center of the footprint. To address this, Figure 14 tracks the radial offset of the peak intensity relative to the overall spray diameter at each height and shows the radial distance of the peak intensity from the spray center, $r_{I_{\max}}$, normalized by the local spray diameter d_z . Values near zero indicate that the densest part of the plume lies close to the spray centerline. Values near one indicate that the energetic core hugs the outer edge of the footprint. In this sense, Figure 14 summarizes how off center the high-density zone is as the spray travels downstream and how this behavior depends on nozzle RPM and propeller configuration.

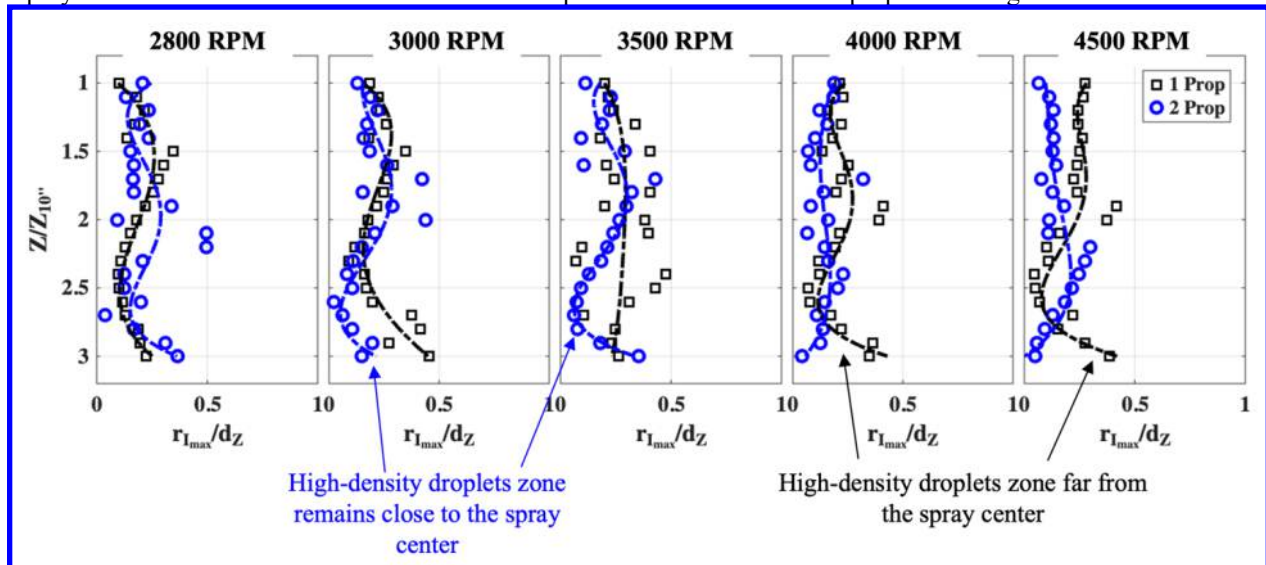


Figure 14 Normalized radial position of maximum spray intensity versus height, showing edge-hugging cores for single propeller and more centered cores for dual propellers.

For the single-propeller configuration, $r_{l_{\max}}/d_z$ is consistently larger than in the dual propeller case across almost all nozzle RPMs and heights. The peak intensity tends to sit closer to the outer edge of the spray, especially at higher RPM, and often migrates outward with increasing $Z/Z_{10''}$. This behavior is consistent with the lopsided, edge-hugging cores seen in the 3D reconstructions and cross sections in Figures 8 and 9 and with the wandering peak locations in Figure 13. As the plume convects downward under a single propeller, the energetic core not only breaks up and wanders but also resides in a region where droplets are more exposed to the surrounding flow and is more sensitive to drift.

The dual propeller configuration shows a different pattern. Across RPMs, $r_{l_{\max}}/d_z$ remains smaller and varies more gently with height. The high-density region stays closer to the interior of the spray footprint rather than sliding toward the outer edge. This agrees with Figure 13, where the trajectory of maximum intensity follows a smooth, repeatable path. Taken together, these results indicate that the second propeller not only stabilizes the path of the energetic core but also anchors it nearer the center of a wider spray column. From an application perspective, this means that in the dual propeller case the bulk of the deposition-relevant droplets occupy an interior band that is less exposed to drift and more likely to produce uniform coverage when adjacent swaths overlap.

IV. Conclusions

This study experimentally investigated how propeller wakes reshape the spray from a Micronair rotary atomizer. Spray volumes were generated at several nozzle RPMs and exposed to either a single propeller located above the atomizer or a dual propeller configuration. A laser patternator and intensity-based image processing were used to reconstruct the three-dimensional spray geometry and to track how droplet concentration, spray diameter, and the energetic core evolved with distance from the nozzle.

The results showed that a single propeller contracts the natural umbrella-shaped plume into a narrower column but also causes the high-intensity core to erode rapidly as the spray convects downstream. The dense region fragments into irregular, lopsided structures and gives way to a broad low-density halo that dominates the spray further downstream from the nozzle. In contrast, the dual propeller configuration produces a stronger and more uniform downwash that confines the plume while preserving a broad, nearly cylindrical high-intensity core with more circular boundaries and a smaller dead zone. Intensity threshold analysis demonstrated that the dual propeller case maintains larger vertically averaged diameters, smaller separation between high and low threshold contours, and a smoother, more repeatable path of maximum intensity that remains closer to the interior of the spray footprint.

From these observations, the most important takeaways are:

- A single propeller narrows the plume but destabilizes the high-intensity core, which collapses and migrates toward the outer edge of the spray footprint.
- A dual propeller wake stabilizes the spray, preserves a wide high-intensity core over the full measured height, and keeps this energetic region closer to the interior of the plume.
- The dual propeller configuration reduces the relative thickness of the diffuse, low-density halo and yields a more coherent and repeatable spray geometry across nozzle RPMs.

These findings suggest that multi-propeller flow fields can be used not only to contract and steer rotary atomizer sprays but also to improve their deposition characteristics. A dual propeller arrangement produces a wider and more coherent column of deposition-relevant droplets, reduces lateral wandering of the core, and limits the growth of highly diffused, drift-prone regions at the periphery. For agricultural applications this behavior points to the potential for more uniform canopy coverage and reduced off-target transport when rotary atomizers are integrated with carefully designed multi-propeller systems.

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